Alireza **Montajab**

Electrical Engineering | Digital Electronics

Personal Information

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Education

2023–2027 (expected) **Bachelor of Science in Electrical Engineering**, *Shahid Beheshti University*, Tehran, Iran
Specialisation: Digital Electronics (from February 2026)
GPA: 3.30/4.0 (16.50/20)
QS World University Rankings 2026: #741–750 worldwide, #214 in Asia

Research Interests

Embedded systems and real-time computing. RTOS scheduling, bare-metal firmware, hardware and software co-design.

Digital and VLSI design. Datapath and logic architecture, CMOS integrated circuits, FPGA prototyping.

Control systems and robotics. Feedback control, state estimation and sensor fusion, mechatronic platforms.

Signal processing and sensing. Digital signal processing for measurement and monitoring, sensor interfacing, vision-based feedback.

Experiences

Work Experience

2026–present **Embedded Hardware Engineer**, *Tensor Intelligent Machines*, Tehran, Iran
Firmware and PCB design for robotic systems at Sharif University Science & Technology Park. Diagnosis and repair of faulty control boards on robots returned from the field.

2025–2026 **Engineering Intern**, *Tensor Intelligent Machines*, Tehran, Iran
Embedded firmware and PCB design; converted to a full-time engineering role.

Teaching Experience

Altium Designer PCB Design Workshop, *Instructor*, Mechatronics Student Association, Shahid Beheshti University

Electrical Circuits, *Teaching Assistant*, Shahid Beheshti University

Projects

Industry Projects

2026 **Handheld Barcode Terminal**, *Tensor Intelligent Machines*
ESP32-S3 firmware using the USB Host HID stack and FreeRTOS, with a custom carrier PCB.

2025 **RFID Reader Board**, *Tensor Intelligent Machines*
Surface-mount reader board designed in Altium Designer, with a fully sourced bill of materials.

2025 **Ultrasonic Ranging Module**, *Tensor Intelligent Machines*
Microsecond-resolution distance measurement on an STM32G0 using hardware timer capture.

Academic Projects

Spring 2026 **Magnetic Levitation System with Visual Feedback**, *Fundamentals of Mechatronics*, Supervisor: Dr. Nouri-Manzar
Stabilised an inherently unstable maglev system using an ESP32-CAM as the position sensor and an on-board PID controller.

Fall 2025 **Water Level and Temperature Control System**, *Digital Systems II*, Supervisor: Dr. Asharioun
Bare-metal STM32 firmware with two control loops, timer-capture ultrasonic sensing and an LCD/keypad menu. Led a team of three.

Spring 2025 **8-Bit Arithmetic Datapath**, *Digital Systems I*, Supervisor: Dr. Farhad Pouladi
Designed an 8-bit datapath from discrete 74LS logic: adders, multiplexers, a decoder and shift registers.

Fall 2025 **Wi-Fi Controlled Smart Lighting**, *Digital Systems II*
ESP32 web server with REST endpoints controlling four lighting channels.

Spring 2025 **H-Bridge Motor Driver PCB**, *Electronics II*
Four-transistor bidirectional DC motor driver designed in Altium Designer.

Fall 2025 **Transmission Line Parameter Calculator**, *Power System Analysis I*
MATLAB tool computing inductance and capacitance of bundled three-phase lines.

Spring 2024 **Op-Amp RLC Network Analysis**, *Electrical Circuits I*
Closed-form transient and steady-state analysis, verified in LTspice.

In Progress

2026–2027 **Digital Signal Processing for Energy Consumption Monitoring**, *Senior Design Project*
Signal acquisition and digital processing for monitoring of electrical energy consumption.

2026–present **Differential Drive Robot with Sensor Fusion**, *Independent Project*
State estimation fusing wheel odometry with an IMU through a Kalman filter.

Selected Courses

A+ CMOS Integrated Circuit Design • Digital Signal Processing • Power System Analysis II

A Digital Systems II • Digital Systems Laboratory I • Principles of Communication Systems • Fundamentals of Mechatronics

Technical Skills

Embedded C, C++, STM32 (HAL, CubeMX), ESP32 / ESP32-S3 (ESP-IDF), FreeRTOS, ARM Cortex-M, bare-metal firmware

Interfaces USB HID, UART, I^2C, SPI, PWM, ADC, DMA, timers, input capture, interrupts

Hardware Altium Designer, SMD PCB design, BOM sourcing, MOSFET power stages, motor drivers, sensor interfacing

Digital Design 74-series logic, finite state machines, datapath design

Control PID design and tuning, closed-loop feedback, real-time image processing

Software Altium Designer, LTspice, Proteus, MATLAB / Simulink, CodeVisionAVR, Arduino IDE, Fritzing, Git

Instruments Oscilloscope, logic analyser, function generator, bench power supply

Activities

2025 Organising committee member, 22nd International Conference of the Iranian Society of Cryptology on Information Security and Cryptology, and of the departmental Electrical Engineering conference, Shahid Beheshti University

Member, Scientific Association of Electrical Engineering, Shahid Beheshti University

Conducted and published an alumni interview for the IEEE Student Branch newsletter, Shahid Beheshti University

Awards and Honors

- Awarded a full tuition scholarship at Shahid Beheshti University (2023–2027), granted on national entrance examination rank.
- Ranked within the top 1% of the Iranian University Entrance Exam (2023).
- High School Diploma, Allameh Tabatabai High School, Tehran (2023), GPA 4.0/4.0.

Languages

English IELTS Academic · Overall 7.0 • Listening 8.0 • Reading 7.5 • Speaking 7.0 • Writing 6.5

Persian Native